Compliance Catch: Auditing 100% of Customer Calls with Agentic AI for Resource-Constrained Nigerian Financial Institutions

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Track: AI research to deployment · Contextual and applied AI

Reviewing every call instead of a 5% sample caught all 8 labelled violations the sample missed, with each flag quoted verbatim and tied to the rule it breaks, for roughly a tenth of a US cent per call.

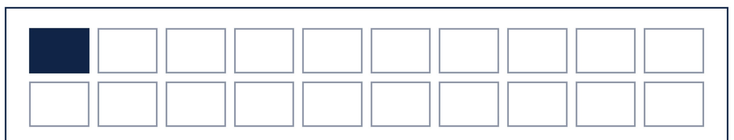
100% of calls reviewed

8 / 8 violations caught (recall 1.0)

0 / 20 hallucinated quotes reached a reviewer

1. Introduction

Nigerian microfinance banks and fintechs face growing consumer-protection and conduct obligations, yet most still rely on manual quality assurance that listens to roughly 5% of customer calls. Illegal guarantees, undisclosed fees, pressure tactics, skipped identity checks and mishandled vulnerable customers go undetected in the other 95%.



The status quo: 1 call in 20 is ever heard by a compliance reviewer. Enterprise call-monitoring platforms close that gap for large banks, but are priced and engineered beyond the reach of smaller institutions.

Research question

Can a small LLM agent review every call, beat a simple keyword scanner on both misses and false alarms, and hand a human reviewer only flags that are already quoted and rule-cited?

2. Materials and Methods

Architecture. A two-pass agent. The second pass is the differentiator: two independent checks, not one model grading itself.

Input
Call audio transcribed to numbered, speaker-labelled lines, plus a plain-language policy library (12 rules).

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Pass 1 · Flagger
Sees the full transcript and full policy. Proposes candidate flags: line, verbatim quote, rule, reasoning, severity.

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Pass 2 · Verifier (two independent checks)

a. Quote check (plain Python, no LLM). Does the quote exist verbatim at that line? If not, straight to *needs review*.

b. Scoped re-check (narrow LLM call). Shown only the one line and the one rule. Yes marks it *grounded*; no downgrades it.

↓

Router and cross-call analysis
Builds the per-call report and aggregates grounded flags across calls to surface recurring rules, agents or scripts.

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Human compliance reviewer
Confirms or dismisses grounded flags. "Grounded" means source-cited and independently re-checked, not legally adjudicated.

Evaluation design

Test set: 10 held-out call transcripts, 8 with one labelled violation, 2 clean; 12-rule policy library.

Baselines: random 5% sample (the status quo) and a keyword/regex scanner reviewing 100% of calls.

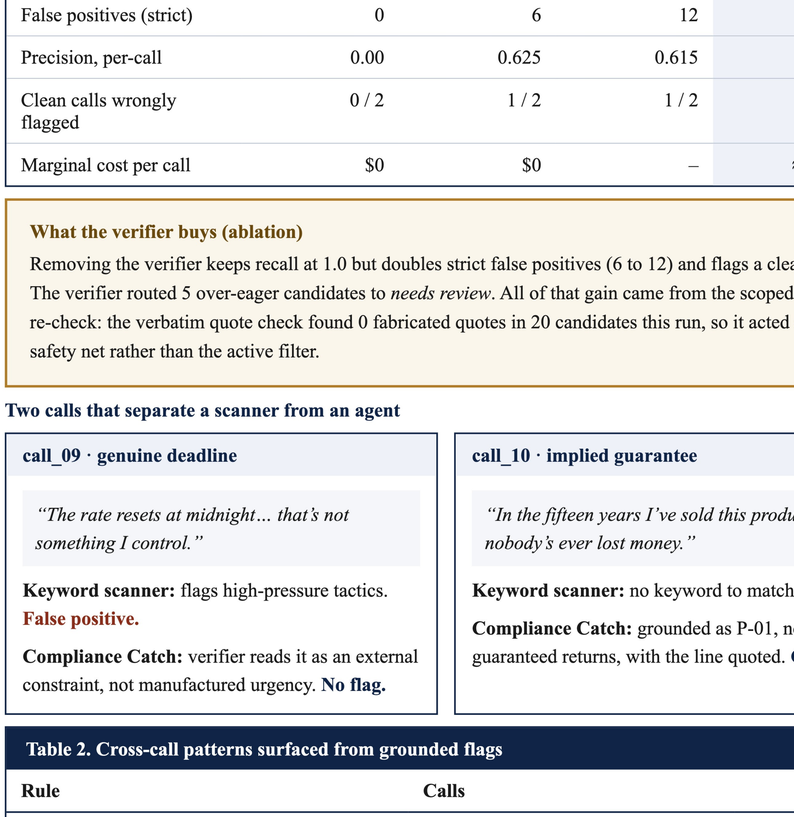
Ablation: flagger only, with the verifier removed.

Scoring: recall and precision under two conventions, per-call (right rule on the right call) and strict (exact line and rule).

Models: gemini-2.5-flash (flagger), gemini-flash-lite-latest (verifier). Every prompt and response is logged as an auditable trajectory.

3. Results and Discussion

Figure 1. Violations caught and strict precision, same 10 calls



Per-call precision (right rule somewhere on a violation call): keyword 0.625, flagger only 0.615, Compliance Catch 0.80. All numbers are emitted by the evaluation script, none hand-computed.

Table 1. Full comparison on 10 held-out calls

Metric	Random 5%	Keyword	Flagger only	Compliance Catch
Coverage	10%	100%	100%	**100%**
Violations caught	0 / 8	5 / 8	8 / 8	**8 / 8**
False positives (strict)	0	6	12	**6**
Precision, per-call	0.00	0.625	0.615	**0.800**
Clean calls wrongly flagged	0 / 2	1 / 2	1 / 2	**0 / 2**
Marginal cost per call	$0	$0	–	**≈ $0.0013**

What the verifier buys (ablation)

Removing the verifier keeps recall at 1.0 but doubles strict false positives (6 to 12) and flags a clean call. The verifier routed 5 over-eager candidates to *needs review*. All of that gain came from the scoped policy re-check: the verbatim quote check found 0 fabricated quotes in 20 candidates this run, so it acted as a safety net rather than the active filter.

Two calls that separate a scanner from an agent

call_09 · genuine deadline

"The rate resets at midnight… that's not something I control."

Keyword scanner: flags high-pressure tactics. **False positive.**

Compliance Catch: verifier reads it as an external constraint, not manufactured urgency. **No flag.**

call_10 · implied guarantee

"In the fifteen years I've sold this product, nobody's ever lost money."

Keyword scanner: no keyword to match. **Missed.**

Compliance Catch: grounded as P-01, no guaranteed returns, with the line quoted. **Caught.**

Table 2. Cross-call patterns surfaced from grounded flags

Rule	Calls	Count
P-01 No guaranteed returns	call_01, call_05, call_10	3
P-07 Accurate risk representation	call_05, call_10	2

A rule recurring across calls points to a script or product problem rather than one agent's slip. A 5% sample would see at most one of these calls and could never show the pattern.

New: first Nigerian Pidgin check (qualitative, unscored)

Setup. Three synthetic Nigerian-context calls: a loan-app collections call in heavy Pidgin, an SME loan-terms call with light code-switching, and a clean Pidgin collections call as a control.

9 / 9 planted violations surfaced by the flagger

6 / 9 grounded; no grounded flag was wrong

0 flags raised on the clean control call

"If you no pay this one by 6pm today, we go send your name and picture to everybody for your contact list say you be debtor."

Grounded as P-11, no unauthorised collections threats, with the line quoted verbatim.

The other 3 went to needs review, not grounded. Each only shows up when you read more than one line: a 9% fee disclosed after the customer had already said "I accept", an interest question dodged, and a complaint brushed off on the next line. The verifier reads one line at a time, so it cannot see that. The flags still reach a human.

Not part of the scored eval. Transcripts and trajectories are in the repo under data/transcripts/nigeria/.

Discussion: three findings

1. **Integrity and judgement are different failures.** The quote check guards against fabricated citations; the scoped re-check guards against over-reading a real line. On this run every error was a judgement error, so the second check did all the work. Both stay: a hallucinated quote reaching a regulator is the failure you cannot afford even once.

2. **Strict precision understates the agent.** All 6 strict false positives sit on two calls (05 and 10) where the agent also grounded a second reinforcing line, or a second overlapping rule ("you can't lose money" is both an implied guarantee and a risk misrepresentation). Each is verbatim and defensible; the single-label answer key simply cannot credit it. Both clean calls were left untouched.

3. **Cost is not the barrier; data governance is.** At about $0.0013 per call the economics work for a microfinance bank. The real preconditions for Nigerian deployment are transcription quality on narrowband audio with English, Pidgin, Yoruba, Hausa and Igbo code-switching, and keeping customer PII inside an approved boundary.

Limits of this evidence

- Scored set is 10 hand-authored English transcripts. Two real contact-centre calls (AxonData, one as audio, one as text) gave the same verdicts by either route, but were not scored.
- Violations by omission (a disclosure never made) have no line to quote and are out of scope for this architecture.
- Reviewer time saved was not measured. Cost is a character-count estimate, not logged tokens.

4. Conclusion and Recommendations

A two-pass agent with an independent verifier reviewed every call, caught every labelled violation, produced fewer false alarms than a keyword scanner, and never passed a fabricated quote to a reviewer. The verifier, not the model size, is what made the output trustworthy.

For institutions adopting this

1. Keep rules in plain language, owned by compliance, so the library tracks CBN and FCCPC guidance without engineers.
2. Never ship a flag without a verbatim quote and a rule citation; make the verifier mandatory.
3. Use the cross-call table as the weekly management view; it is where sampling fails most.
4. Swap in a local or cheaper model where data residency or budget demands; the pipeline is provider-agnostic.

Next steps

Multi-label ground truth; extend the first Pidgin set and add reference transcripts; give the verifier the surrounding lines so violations that span two lines (for example, a fee mentioned after commitment) can be grounded; a timed reviewer study with a microfinance partner; presence-mode rules for omissions.

Code, prompts, trajectories and evaluation script
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External validation audio: AxonData english-contact-center-audio-dataset (CC BY-NC 4.0), not redistributed